Angular's Build Architechture changes from Angular 2 to Latest

🧱 Angular Build Architecture Evolution

Angular Version	Build Tool / Architecture	Compiler	Key Features / Notes
2.x – 4.x (2016–2017)	**SystemJS / Webpack (manual setup)**	**View Engine (JIT / AOT)**	- Used @angular/compiler for JIT.- Optional AOT (Ahead-of-Time) compiler.- Manual Webpack or SystemJS setups.- No CLI bundler yet.
5.x – 7.x (2017–2018)	**Angular CLI + Webpack (via @angular-devkit/build-angular)**	**View Engine (AOT default)**	- Angular CLI standardizes builds.- Introduced angular.json for build configuration.- Builds used Webpack internally.- Differential loading (modern + legacy JS).
8.x (2019)	**Angular CLI + Webpack (View Engine + Ivy preview)**	**Ivy (opt-in)**	- Ivy compiler introduced as preview.- Still uses Webpack-based CLI builder.- Faster incremental builds (still early).
9.x – 11.x (2020–2021)	**Webpack (Ivy default)**	**Ivy (default)**	- Ivy replaces View Engine.- Better tree-shaking & build speed.- Introduced ngcc (Angular compatibility compiler) to convert old View Engine packages.- @angular-devkit/build-angular main builder.
12.x – 14.x (2021–2022)	**Webpack 5 + Angular CLI**	**Ivy**	- Webpack 5 becomes default.- Deprecated ngcc pipeline for future.- Introduced **esbuild** experimentation internally.- Faster builds, persistent caching.- Angular Package Format (APF v13+).
15.x (2022)	**Webpack 5 (stable) + Optional esbuild preview**	**Ivy + Standalone Components**	- Standalone components reduce NgModule dependency.- Partial builds with esbuild experiments begin.- Angular CLI still Webpack-driven.

16.x (2023)	**Hybrid Build: Webpack + esbuild (Preview)**	**Ivy + Signals (developer preview)**	- Major re-architecture prep.- Introduced @angular/build experimental esbuild-based builder.- esbuild & Vite integration experiments.- Beginning of future replacement for Webpack.
17.x (Nov 2023)	**New build system: esbuild + Vite (default)**	**Ivy / JIT compatible**	- **Goodbye Webpack by default.**- @angular/build replaces @angular-devkit/build-angular.- Vite + esbuild for dev server + bundling.- 90% faster HMR & rebuilds.- CLI simplified configuration.- Hybrid SSR support.
18.x (May 2024)	**Vite + esbuild (stable)**	**Ivy (optimized)**	- Fully stable new builder (application, browser, dev-server).- Simplified angular.json structure.- Universal SSR upgraded (based on Vite middleware).- ng build and ng serve now powered by esbuild + Vite.
19.x (Nov 2024 – 2025)	**Vite + esbuild (mature)**	**Ivy (final) / Preparations for Angular "Next" compiler**	- Unified build + SSR + prerender architecture.- New "outputMode": "server" & "renderMode" APIs.- "@angular/ssr" fully integrated.- @angular/build is now official and stable.- Angular CLI config simplified for hybrid rendering.

🧩 Summary of Major Architectural Milestones

Era	Core Builder	Key Concept	Notes
Angular 2–4	Manual (SystemJS/Webpack)	View Engine	Early, non-standardized builds.
Angular 5–8	Angular CLI + Webpack	View Engine + ngc	AOT compilation becomes mainstream.
Angular 9–14	Angular CLI + Webpack 5	Ivy Compiler	Ivy brings major rendering + build improvements.
Angular 15–16	Webpack + esbuild (Preview)	Hybrid builds	Transition to faster toolchains.

Angular 17–19	Vite + esbuild (default)	Modern Build System	CLI simplified, HMR, SSR, hybrid rendering.

⚙️ Config Changes in angular.json (simplified overview)

Version	Builder Example	Notes
v8–v16	@angular-devkit/build-angular:browser, @angular-devkit/build-angular:dev-server	Webpack-based
v17+	@angular/build:application, @angular/build:browser, @angular/build:dev-server	esbuild + Vite based
v18+	Added "ssr" and "prerender" targets	Unified build + render modes
v19	outputMode: "server", renderMode: RenderMode.Hybrid	Hybrid SSR + CSR rendering architecture

🚀 Key Takeaways

- **Angular 2–8:** Webpack manually or via CLI → slow rebuilds.
- **Angular 9–14:** Ivy compiler → smaller, faster, cleaner builds.
- **Angular 17+:** Complete rewrite of the build pipeline with **esbuild + Vite**.
- **Angular 19:** SSR and client builds unified under a single builder — **@angular/build**.